Lemma 5.4.4: *Let $(X, \precsim)$ be a partially ordered set. An element a is a minimal element in X iff $(\forall x \in X)(x \precsim a \Rightarrow x = a)$.*

Proof: a is a minimal element in $X \Leftrightarrow \sim (\exists x \in X) x \prec a$ (Definition of minimal element)

$\Leftrightarrow (\forall x \in X) \sim (x \prec a)$ (FE8$_\in$)

$\Leftrightarrow (\forall x \in X) \sim (x, a) \in (\precsim - X_=)$ (Definition of $\prec$)

$\Leftrightarrow (\forall x \in X) \sim ((x, a) \in \precsim \wedge (x, a) \notin X_=)$ (Definition of $-$)

$\Leftrightarrow (\forall x \in X) \sim (x \precsim a \wedge x \neq a)$ (equivalent notation)

$\Leftrightarrow (\forall x \in X)(\sim (x \precsim a) \vee x = a)$ (E16, E15)

$\Leftrightarrow (\forall x \in X)((x \precsim a) \Rightarrow x = a)$ (E18) ■

5.5 Well Ordering

Definition: A partially ordered set $(X, \precsim)$ is a ***well-ordered set*** and $\precsim$ is a ***well ordering*** if every non-empty subset of X has a smallest element.

Let $\mathbf{N}_0$ be the set of non-negative integers. i.e. $\mathbf{N}_0 = \mathbf{N} \cup \{0\}$, where $\mathbf{N}$ is the set of pisitive integers.

The well ordering principle of nonnegative integers (WOP)

Every non-empty set of nonnegative integers has a smallest element. Formally,

$$(\forall X)((X \subseteq \mathbf{N}_0 \wedge X \neq \emptyset) \Rightarrow (\exists x \in X)(\forall y)(y \in X \Rightarrow x \leq y))$$

Mathematical Induction

Mathematical induction is a powerful tool for proving theorems of the form "every nonnegative integer possesses property P". It will be used in Part III to prove theorems related to graphs. Mathematical induction is also used extensively in proving the correctness of algorithms. In the following, we shall use the well-ordering principle of nonnegative integers to prove the correctness of mathematical induction.

There are two versions of mathematical induction: the Principle of (ordinary) mathematical induction and the Principle of strong mathematical induction.

5.5.1.1 The Principle of (ordinary) Mathematical Induction

Let $P(n)$ be a predicate on nonnegative integers n.

If $\begin{cases} P(0) \equiv true, \text{ and} \\ P(k) \equiv true \Rightarrow P(k+1) \equiv true, \text{for every nonnegative integer } k, \end{cases}$

then $P(n) \equiv true$ for every nonnegative integer n.

Rewriting the principle as an inference rule, we have:

PMI: $\dfrac{P(0), (\forall k \in \mathbf{N}_0)(P(k) \Rightarrow P(k+1))}{(\forall n \in \mathbf{N}_0) P(n)}$, where $\mathbf{N}_0$ is the set of non–negative integers.

We shall prove the correctness of **PMI** using the well ordering principle of nonnegative integers **WOP**. Specifically,

given $S \subseteq \mathbf{N}_0$ such that $0 \in S$ and $\forall n \in \mathbf{N}_0, n \in S \Rightarrow n + 1 \in S$. We are to prove that $S = \mathbf{N}_0$.

Proof: (Proof by contradiction) Suppose $S \neq \mathbf{N}_0$. ..... (A)

By the definition of S, we have: $S \subseteq \mathbf{N}_0$, ····· (I)

$0 \in S$, ····· (II)

$\forall n \in \mathbf{N}_0, n \in S \Rightarrow n+1 \in S$. ····· (III)

Let $T = \mathbf{N}_0 - S$ ····· (B)

$S \subseteq \mathbf{N}_0 \wedge S \neq \mathbf{N}_0$ ((I),(A),I2)

$\Rightarrow S \subset \mathbf{N}_0$ (Definition of $\subset$)

$\Rightarrow (\exists x)(x \in \mathbf{N}_0 \wedge x \notin S)$ (Lemma 3.2.6)

$\Rightarrow (\exists x)(x \in \mathbf{N}_0 - S)$ (Definition of $-$)

$\Rightarrow \mathbf{N}_0 - S \neq \emptyset$ (Lemma 3.5.8)

$\Rightarrow T \neq \emptyset$ ((B),Sub$_=$) ····· (IV)

$\mathbf{N}_0 \cap \overline{S} \subseteq \mathbf{N}_0$ (Corollary 4.2.2.1)

$\Rightarrow \mathbf{N}_0 - S \subseteq \mathbf{N}_0$ (Theorem 4.3.7(v))

$\Rightarrow T \subseteq \mathbf{N}_0$ ((B),Sub$_=$) ····· (V)

$\Rightarrow (T \subseteq \mathbf{N}_0) \wedge (T \neq \emptyset)$ ((V),(IV),I2) ····· (VI)

$(\forall X)((X \subseteq \mathbf{N}_0 \wedge X \neq \emptyset) \Rightarrow (\exists x \in X)(\forall y)(y \in X \Rightarrow x \leq y))$ (WOP)

$\Rightarrow (T \subseteq \mathbf{N}_0 \wedge T \neq \emptyset) \Rightarrow (\exists x \in T)(\forall y)(y \in T \Rightarrow x \leq y)$ (UI) ····· (VII)

$\Rightarrow (\exists x \in T)(\forall y)(y \in T \Rightarrow x \leq y)$ ((VI),(VII),I3)

$\Rightarrow m \in T \wedge (\forall y)(y \in T \Rightarrow m \leq y)$ (EI)

$$\Rightarrow \begin{cases} m \in T \\ (\forall y)(y \in T \Rightarrow m \leq y) \end{cases}$$ (E9,I2)

$$\Rightarrow \begin{cases} m \in \mathbf{N}_0 - S & \cdots\cdots \text{(VIII)} \\ (\forall y)(y \in \mathbf{N}_0 - S \Rightarrow m \leq y) & \cdots\cdots \text{(IX)} \end{cases}$$ ((B),Sub$_=$)

$\Rightarrow m \in \mathbf{N}_0 \wedge m \notin S$ ((VIII), Definition of $-$)

$$\Rightarrow \begin{cases} m \in \mathbf{N}_0 & \cdots\cdots \text{(X)} \\ m \notin S & \cdots\cdots \text{(XI)} \end{cases}$$ (E9,I2)

If $m = 0$, then $m \in S$ ((II),Sub$_=$) ····· (XII)

$\Rightarrow m \in S \wedge m \notin S$, a contradiction! ((XII),(XI),I2)

Hence, $m \neq 0$

$\Rightarrow m \geq 1$ (Property of $\mathbf{N}_0$)

$\Rightarrow m - 1 \geq 0$ (Property of $\mathbf{N}_0$)

$\Rightarrow m - 1 \in \mathbf{N}_0$ (Property of $\mathbf{N}_0$) ····· (XIII)

Now, $m - 1 \in \mathbf{N}_0 - S \Rightarrow m \leq m - 1$ ((IX),UI) ····· (XIV)

and $m \nleq m - 1$ (Property of $\mathbf{N}_0$) ····· (XV)

$\Rightarrow m - 1 \notin \mathbf{N}_0 - S$ ((XV),(XIV),I4)

$\Rightarrow \sim (m - 1 \in \mathbf{N}_0 \wedge m - 1 \notin S)$ (Definition of $-$)

$\Rightarrow m - 1 \in \mathbf{N}_0 \Rightarrow m - 1 \in S$ (E16,E15,E18) ····· (XVI)

$\Rightarrow m - 1 \in S$ ((XIII),(XVI),I3) ····· (XVII)

Hence, $(m-1) \in \mathbf{N}_0 \Rightarrow ((m-1) \in S \Rightarrow (m-1)+1 \in S)$ ((III),UI) ····· (XVIII)

$\Rightarrow (m-1) \in S \Rightarrow (m-1)+1 \in S$ ((XIII),(XVIII),I3) ····· (XIX)

$\Rightarrow (m-1)+1 \in S$ ((XVII),(XIX),I3)

$\Rightarrow m \in S$ ..... (XX)

$\Rightarrow m \in S \wedge m \notin S$, a contradiction! ((XX),(XI),I2) ■

The Principle of Strong Mathematical Induction

Let $P(n)$ be a predicate on nonnegative integers n.

If $\begin{cases} P(0) \equiv true, \text{ and} \\ (\forall m \leq k)(P(m) \equiv true) \Rightarrow P(k+1) \equiv true, \text{for every nonnegative integer } k, \end{cases}$

then $P(n) \equiv true$ for every nonnegative integer n.

Rewriting the principle as an inference rule, we have:

PSMI: $\dfrac{P(0), (\forall k \in \mathbf{N}_0)((\forall m \in \mathbf{N}_0)(m \leq k \Rightarrow P(m)) \Rightarrow P(k+1))}{(\forall k \in \mathbf{N}_0)P(n)}$.

As with **PSM**, the correctness of **PSMI** can be proved based on **WOP**.

5.6 Operations on Relations

Since relations are sets, set operators such as union, intersection, relative and absolute complements, and Cartesian product can be naturally applied to relations. In this section, we shall introduce two other operations which are applicable to relations only.

Definition: Let R be a relation in a set X. The ***inverse*** of R is the relation $R^{-1} = \{(y, x) \mid (x, y) \in R\}$

Example 5.6.16

(i) Let $R = \{(1,2), (1,3), (a,b), (\alpha, \beta)\}$. Then $R^{-1} = \{(2,1), (3,1), (b,a), (\beta, \alpha)\}$.

(ii) The *marriage* relation M can be defined as $M = \{(x,y) \mid x \, is \, the \, husband \, of \, y\}$. Then $M^{-1} = \{(y,x) \mid y \, is \, the \, wife \, of \, x\} = \{(x,y) \mid x \, is \, the \, wife \, of \, y\}$.

(iii) $(X \times Y)^{-1} = Y \times X$.

(iv) The inverse of $\leq = \{(x,y) \in \mathbf{R} \times \mathbf{R} \mid x \leq y\}$ is the relation $\leq^{-1} = \{(y,x) \in \mathbf{R} \times \mathbf{R} \mid x \leq y\} = \{(x,y) \in \mathbf{R} \times \mathbf{R} \mid x \geq y\}$, where $\mathbf{R}$ is the set of all real numbers. Specifically, the inverse of the *less-than-or-equal* relation between real numbers is the *greater-than-or-equal* relation. The relation $\leq^{-1}$ is customary written as $\geq$. In general, the inverse of a partial order $\preceq$ is denoted by $\succeq$.

(v) The *divisibility* relation is the relation: $| = \{(x,y) \in \mathbf{Z} \times \mathbf{Z} \mid x \, divides \, y\}$. The inverse of $|$ is the relation $|^{-1} = \{(x,y) \in \mathbf{Z} \times \mathbf{Z} \mid x \, is \, a \, multiple \, of \, y\}$. ■

Definition: Let $R_1 \subseteq A \times B$ and $R_2 \subseteq B \times C$. The ***composition*** of R_1 and R_2 is the relation $R_1 \circ R_2 = \{(x,z) \in A \times C \mid (\exists y \in B)((x,y) \in R_1 \wedge (y,z) \in R_2)\}$.

Example 5.6.17

(i) Let P be a set of patients, T be a set of treatments and S be a set of symptoms.

Let $R = \{(p,s) \in P \times S \mid p \, has \, sympton \, s\}$, and

$S = \{(s,t) \in S \times T \mid sympton \, s \, needs \, treatment \, t\}$

Then $R \circ S = \{(p,t) \in P \times T \mid patient \, p \, needs \, treatment \, t\}$.

(ii) Let $R_1 = \{(\alpha, a), (\beta, a), (\gamma, b)\}$, and

$R_2 = \{(a, \alpha), (b, \beta), (b, \alpha), (c, \gamma)\}$.

Then $R_1 \circ R_2 = \{(\alpha,\alpha),(\beta,\alpha),(\gamma,\beta),(\gamma,\alpha)\}$ and $R_2 \circ R_1 = \{(a,a),(b,a),(c,b)\}$. ■

Theorem 5.6.5: *Let R, S and T be relations.*

(i) $\left(R^{-1}\right)^{-1} = R$.

(ii) $\circ$ *is not commutative.*

(iii) $(R \circ S) \circ T = R \circ (S \circ T)$ *where* $R \subseteq X \times Y$, $S \subseteq Y \times Z$, $T \subseteq Z \times W$.

(iv) $(R \circ S)^{-1} = S^{-1} \circ R^{-1}$.

(v) $R \subseteq S \Rightarrow R^{-1} \subseteq S^{-1}$.

Proof:

(i) (*Direct proof*)

$$\begin{aligned}(x,y) \in \left(R^{-1}\right)^{-1} &\Leftrightarrow (y,x) \in \left(R^{-1}\right) \quad \left(\text{Definition of } R^{-1}\right) \\ &\Leftrightarrow (x,y) \in R \quad \left(\text{Definition of } R^{-1}\right)\end{aligned}$$

Therefore, $(\forall u)\left(u \in \left(R^{-1}\right)^{-1} \Leftrightarrow u \in R\right)$ (Gen)

Hence, $\left(R^{-1}\right)^{-1} = R$ (Principle of Extension) □

(ii) (*Proof by Counterexample*)

Let $R = \{(a,1)\}$ and $S = \{(1,b)\}$

Then $R \circ S = \{(a,b)\}$ but $S \circ R = \emptyset$.

Therefore, $R \circ S \neq S \circ R$. □

(iii) (Bidirectional Proof)

$$\begin{aligned}&(a,b) \in (R \circ S) \circ T \\ \Leftrightarrow\ &(\exists z)(z \in Z \wedge ((a,z) \in (R \circ S) \wedge (z,b) \in T)) \quad \text{(Definition of } \circ) \\ \Leftrightarrow\ &(\exists z)(z \in Z \wedge ((\exists y)(y \in Y \wedge ((a,y) \in R \wedge (y,z) \in S)) \wedge (z,b) \in T)) \quad \text{(Definition of } \circ) \\ \Leftrightarrow\ &(\exists z)((\exists y)(y \in Y \wedge ((a,y) \in R \wedge (y,z) \in S)) \wedge z \in Z \wedge (z,b) \in T) \quad \text{(E9, E11)} \\ \Leftrightarrow\ &(\exists z)(\exists y)((y \in Y \wedge ((a,y) \in R \wedge (y,z) \in S)) \wedge z \in Z \wedge (z,b) \in T) \\ &\qquad \text{(FE6, } z \in Z \wedge (z,b) \in T \text{ contains no free } y) \\ \Leftrightarrow\ &(\exists y)((y \in Y \wedge ((a,y) \in R \wedge (y,d) \in S)) \wedge d \in Z \wedge (d,b) \in T) \quad (\Rightarrow) \text{EI, } d \text{ is constant; } \Leftarrow) \text{EQ)} \\ \Leftrightarrow\ &(c \in Y \wedge ((a,c) \in R \wedge (c,d) \in S)) \wedge d \in Z \wedge (d,b) \in T \quad (\Rightarrow) \text{EI, } c \text{ is constant; } \Leftarrow) \text{EQ)} \\ \Leftrightarrow\ &(\exists z)(c \in Y \wedge ((a,c) \in R \wedge (c,z) \in S)) \wedge z \in Z \wedge (z,b) \in T \quad (\Rightarrow) \text{EQ; } \Leftarrow) \text{EI, } d \text{ is constant)} \\ \Leftrightarrow\ &(\exists y)(\exists z)((y \in Y \wedge ((a,y) \in R \wedge (y,z) \in S)) \wedge z \in Z \wedge (z,b) \in T) \quad (\Rightarrow) \text{EQ; } \Leftarrow) \text{ EI, } c \text{ is constant)} \\ \Leftrightarrow\ &(\exists y)(\exists z)(z \in Z \wedge ((y,z) \in S \wedge (z,b) \in T) \wedge y \in Y \wedge (a,y) \in R) \quad \text{(E9, E11)} \\ \Leftrightarrow\ &(\exists y)((\exists z)(z \in Z \wedge ((y,z) \in S \wedge (z,b) \in T)) \wedge y \in Y \wedge (a,y) \in R) \\ &\qquad \text{(FE6, } y \in Y \wedge (a,y) \in R \text{ contains no free } z) \\ \Leftrightarrow\ &(\exists y)((y,b) \in S \circ T \wedge y \in Y \wedge (a,y) \in R) \quad \text{(Definition of } \circ) \\ \Leftrightarrow\ &(\exists y)(y \in Y \wedge (a,y) \in R \wedge (y,b) \in S \circ T) \quad \text{(E9, E11)} \\ \Leftrightarrow\ &(a,b) \in R \circ (S \circ T) \qquad \text{(Definition of } \circ)\end{aligned}$$

Therefore, $(\forall u)(u \in (R \circ S) \circ T \Leftrightarrow u \in R \circ (S \circ T))$ (Gen)
Hence, $(R \circ S) \circ T = R \circ (S \circ T)$ (Principle of Extension) □

(iv) (Bidirectional Proof) Let $R \subseteq X \times Y$, $S \subseteq Y \times Z$.

$(a,b) \in (R \circ S)^{-1} \Leftrightarrow (b,a) \in R \circ S$ (Definition of R^{-1})
$\Leftrightarrow (\exists x \in Y)((b,x) \in R \wedge (x,a) \in S)$ (Definition of $\circ$)
$\Leftrightarrow (\exists x \in Y)\big((x,b) \in R^{-1} \wedge (a,x) \in S^{-1}\big)$ (Definition of R^{-1}, S^{-1})
$\Leftrightarrow (\exists x \in Y)\big((a,x) \in S^{-1} \wedge (x,b) \in R^{-1}\big)$ (E9)
$\Leftrightarrow (a,b) \in S^{-1} \circ R^{-1}$ (Definition of $\circ$)

Therefore, $(\forall u)\Big(u \in (R \circ S)^{-1} \Leftrightarrow u \in S^{-1} \circ R^{-1}\Big)$ (Gen)
Hence, $(R \circ S)^{-1} = S^{-1} \circ R^{-1}$. (Principle of Extension) □

(v) (Direct Proof) Suppose $R \subseteq S$.

Then $(x,y) \in R \Rightarrow (x,y) \in S$ . . . . (I)
Let $(a,b) \in R^{-1}$. Then, $(b,a) \in R$ (Definition of R^{-1})
$\Rightarrow (b,a) \in S$ ((I), I3)
$\Rightarrow (a,b) \in S^{-1}$ (Definition of S^{-1})

Therefore, $(\forall u)\big(u \in R^{-1} \Rightarrow u \in S^{-1}\big)$ (Gen)
Hence, $R^{-1} \subseteq S^{-1}$. (Definition of $\subseteq$) ■

5.7 Examples

1. Illustrate the validity of Theorem 5.3.6(i) with the following relation.

R is a relation in $X = \{a,b,c,d,e,f\}$ such that
$R = \{(a,a),(a,b),(b,a),(b,b),(c,c),(d,d),(e,e),(f,f),(c,d),(c,f),(f,d),(d,c),(d,f),(f,c)\}$.

Solution:

By inspection, R is reflexive, symmetric and transitive. Therefore, R is an equivalence relation.

The equivalence classes w.r.t. R are:

$[a]/R = \{a,b\}$
$[b]/R = \{a,b\}$
$[c]/R = \{c,d,f\}$
$[d]/R = \{c,d,f\}$
$[e]/R = \{e\}$
$[f]/R = \{c,d,f\}$

By inspection, $[X]/R = \{\{a,b\},\{c,d,f\},\{e\}\}$ is a partition of X.

The relation induced by $[X]/R$ is the set:
$\{(a,a),(a,b),(b,a),(b,b),(c,c),(d,d),(f,f),(c,d),(c,f),(f,d),(d,c),(d,f),(f,c),(e,e)\}$.
which is identical to the relation R. ■

2. Illustrate the validity of Theorem 5.3.6(ii) with the following set X and partition C.

$$X = \{a, b, c, d, e, f\} \text{ and } \mathcal{C} = \{\{a, b, c\}, \{d\}, \{e, f\}\}.$$

Solution:

The relation in X induced by $\mathcal{C}$ is:

$$X/\mathcal{C} = \{(a,a), (b,b), (c,c), (a,b), (a,c), (b,c), (b,a), (c,a), (c,b), (d,d), (e,e), (f,f), (e,f), (f,e)\}.$$

By inspection, $X/\mathcal{C}$ is reflexive, symmetric and transitive.

Therefore, $X/\mathcal{C}$ is an equivalence relation.

The equivalence classes w.r.t. $X/\mathcal{C}$ are:

$$[a]/(X/\mathcal{C}) = \{a, b, c\}$$
$$[b]/(X/\mathcal{C}) = \{a, b, c\}$$
$$[c]/(X/\mathcal{C}) = \{a, b, c\}$$
$$[d]/(X/\mathcal{C}) = \{d\}$$
$$[e]/(X/\mathcal{C}) = \{e, f\}$$
$$[f]/(X/\mathcal{C}) = \{e, f\}$$

Therefore, the collection of equivalence classes w.r.t. $X/\mathcal{C}$ is $[X]/(X/\mathcal{C}) = \{\{a, b, c\}, \{d\}, \{e, f\}\}$ which is identical to the partition $\mathcal{C}$. ■

3. Let $\mathbf{R}$ be the set of real numbers. Prove that the following relation S is an equivalence relation in $\mathbf{R} \times \mathbf{R}$.

$$S = \{((a, b), (c, d)) \in (\mathbf{R} \times \mathbf{R}) \times (\mathbf{R} \times \mathbf{R}) | ab = cd\}$$

Solution:

(i) To prove that S is reflexive, we shall prove that: $\forall (a, b) \in \mathbf{R} \times \mathbf{R}, ((a, b), (a, b)) \in S$

Let $(a, b) \in \mathbf{R} \times \mathbf{R}$. As $ab = ab$, we have $((a, b), (a, b)) \in S$. (Definition of S)

Hence, S is reflexive.

(ii) To prove that S is symmetric, we shall prove that:

$$\forall (a, b), (c, d) \in \mathbf{R} \times \mathbf{R}, ((a, b), (c, d)) \in S \Rightarrow ((c, d), (a, b)) \in S.$$

Let $((a, b), (c, d)) \in S$ Then $ab = cd$ (Definition of S)

$\Rightarrow cd = ab$ ($\text{Sub}_=$)

$\Rightarrow ((c, d), (a, b)) \in S$ (Definition of S)

Hence, S is symmetric.

(iii) To prove that S is transitive, we shall prove that:

$\forall (a, b), (c, d), (e, f) \in \mathbf{R} \times \mathbf{R},$

$$((a, b), (c, d)) \in S \wedge ((c, d), (e, f)) \in S \Rightarrow ((a, b), (e, f)) \in S$$

Let $((a, b), (c, d)) \in S \wedge ((c, d), (e, f)) \in S$.

Then $(ab = cd) \wedge (cd = ef)$ (Definition of S)

$\Rightarrow ab = cd$ and $cd = ef$ (E9,I2)

$\Rightarrow ab = ef$ ($\text{Sub}_=$)

$\Rightarrow ((a, b), (e, f)) \in S$ (Definition of S)

Hence, S is transitive. ■

4. Prove that R is a partial order in X iff R^{-1} is a partial order in X.

(i) R is reflexive $\Leftrightarrow \forall x \in X, (x,x) \in R$ (Definition of reflexivity)
$\Leftrightarrow \forall x \in X, (x,x) \in R^{-1}$ (Definition of R^{-1})
$\Leftrightarrow R^{-1}$ is reflexive (Definition of reflexivity)

(ii) R is antisymmetric $\Leftrightarrow \forall x, y \in X, (x,y) \in R \wedge (y,x) \in R \Rightarrow (x = y)$
(Definition of antisymmetricity)
$\Leftrightarrow \forall x, y \in X, (y,x) \in R^{-1} \wedge (x,y) \in R^{-1} \Rightarrow (x = y)$
(Definition of R^{-1})
$\Leftrightarrow \forall x, y \in X, (x,y) \in R^{-1} \wedge (y,x) \in R^{-1} \Rightarrow (x = y)$ (E9)
$\Leftrightarrow R^{-1}$ is antisymmetric (Definition of transitivity)

(iii) R is transitive $\Leftrightarrow \forall x, y, z \in X, (x,y) \in R \wedge (y,z) \in R \Rightarrow (x,z) \in R$ (Definition of transitivity)
$\Leftrightarrow \forall x, y, z \in X, (y,x) \in R^{-1} \wedge (z,y) \in R^{-1} \Rightarrow (z,x) \in R^{-1}$
(Definition of R^{-1})
$\Leftrightarrow \forall x, y, z \in X, (z,y) \in R^{-1} \wedge (y,x) \in R^{-1} \Rightarrow (z,x) \in R^{-1}$ (E9)
$\Leftrightarrow R^{-1}$ is transitive. (Definition of transitivity)

Hence, R is a partial order in X iff R^{-1} is a partial order in X. ■

5.8 Exercises

1. Let R be a relation from $A = \{1,2,3,4\}$ to $B = \{1,3,5\}$ such that $(x,y) \in R$ iff $x < y$. Write R as a set of ordered pairs.
2. Prove that if R is a symmetric relation, then $R \cap R^{-1} = R$.
3. Let R and S be relations in a set A. Prove or disprove the following:
   (a) If R and S are asymmetric, then (i) $R \cup S$ is a symmetric relation; (ii) $R \cap S$ is a symmetric relation.
   (b) If R and S are antisymmetric, then (i) $R \cup S$ is an antisymmetric relation; (ii) $R \cap S$ is an antisymmetric relation.
4. Let R_1 and R_2 be two equivalence relations in a set A. Prove that $R_1 \cap R_2$ is an equivalence relation in A, but $R_1 \cup R_2$ need not be an equivalence relation.
5. Let A_1, A_2 and A_3 be non-empty sets and $X = A_1 \cup A_2 \cup A_3$. Prove that the set $\{A_1, A_2 - A_1, A_3 - (A_1 \cup A_2)\} - \{\emptyset\}$ is a partition of X.
6. Use Theorem 5.3.6 to prove that the following relation R is an equivalence relation.
   Let $\mathbf{R}^+$ be the set of nonnegative real numbers and $\lfloor x \rfloor$ is the largest integer less than or equal to x. The relation R is defined as:
   $$R = \{(a,b) \in \mathbf{R}^+ \times \mathbf{R}^+ \mid \lfloor a \rfloor = \lfloor b \rfloor\}$$
7. Let R be a symmetric and transitive relation in a set A. Prove that if for every $a \in A$, there exists $b \in A$ such that $(a,b) \in R$, then R is an equivalence relation. Show that the conclusion is false if symmetricity is replaced by reflexivity.
8. Let $\mathbf{Z}$ be the set of integers and R is a relation in $\mathbf{Z}$ such that
   $$R = \{(a,b) \in \mathbf{Z} \times \mathbf{Z} \mid a^2 - b^2 \text{ is an even integer}\}.$$
   Prove that R is an equivalence relation.
9. Let $\mathbf{N}$ be the set of all positive integers and R is a relation in $\mathbf{N}$ such that
   $$R = \{(a,b) \in \mathbf{N} \times \mathbf{N} \mid \text{the sum of the decimal digits in } a = \text{the sum of the decimal digits in } b\}.$$
   Prove that R is an equivalence relation.

Describe the equivalence class $[98]/R$. List all the integers in $[98]/R$ that fall within the range from 0 to 50, inclusive.

10. Let R be an equivalence relation in X and R' be an equivalence relation in Y. Let $((x,y),(x',y')) \in S$ iff $(x,x') \in R \wedge (y,y') \in R'$. Prove that S is an equivalence relation in $X \times Y$.
11. Let $\precsim$ be a partial order and $\prec$ be its corresponding strict order. Prove that:
    (a) if $x \precsim y$ and $y \prec z$, then $x \prec z$.
    (b) if $x \prec y$ and $y \precsim z$, then $x \prec z$.
12. Let R be a reflexive and transitive relation in X. Let S be a relation in X such that $(x,y) \in S \Leftrightarrow (x,y) \in R \wedge (y,x) \in R$.
    (a) Prove that S is an equivalence relation.
    (b) Let $\widetilde{R}$ be a relation in $[X]/S$ such that $\widetilde{R} = \{([x]/S,[y]/S) \mid (x,y) \in R\}$. Prove that $\widetilde{R}$ is a partial order.
13. Let $(X,\lesssim)$ and $(Y,\preceq)$ be partially ordered sets. Let $(x,y) \sqsubseteq (x',y')$ iff $(x \lesssim x') \wedge (y \preceq y')$. Prove that $(X \times Y, \sqsubseteq)$ is a partially ordered set.
14. Let $\mathcal{C}$ and $\mathcal{C}'$ be partitions of X. $\mathcal{C}$ is *finer than* $\mathcal{C}'$, denoted by $\mathcal{C} \Subset \mathcal{C}'$, iff $S \in \mathcal{C} \Rightarrow (\exists S' \in \mathcal{C}')S \subseteq S'$. Prove that $\Subset$ is a partial order in the set of all partitions of X.
15. Let $(X,\lesssim)$ be a partially ordered set. Prove that if X has two distinct minimal elements, then it has no smallest element.
16. Let R be a relation in X. Prove that $R \cup R^{-1}$ is the smallest symmetric relation including R and that $R \cap R^{-1}$ is the largest symmetric relation included in R.
17. Let R be a relation from X to Y. Let $X_=$ and $Y_=$ be the equality relation in X and Y, respectively. Prove that $R \circ Y_= = R$ and $X_= \circ R = R$.
18. Let R be a relation in X. Prove that:
    (a) R is reflexive iff $X_= \subseteq R$
    (b) R is irreflexive iff $X_= \cap R = \emptyset$
    (c) R is symmetric iff $R = R^{-1}$
    (d) R is antisymmetric iff $R \cap R^{-1} \subseteq X_=$
    (e) R is transitive iff $R \circ R \subseteq R$.
19. Let $\prec$ be a strict order in X. Prove that $(\prec \cup X_=) - X_= = \prec$.
20. Let R be a relation in a set X. If $X \neq \emptyset$, then R is reflexive $\Rightarrow$ R is not asymmetric.